



**Topic:** Design a Maze + Write the Solution · **Course:** Maze Madness · **Level:** Extension (Advanced) · **Time:** about 60 minutes

This time you are the maze maker. Design one maze that **stretches through 6 zones**, wire it with pistons and doors, then write the **solution code** that drives the agent to the end.

Open **M002 EXT 3 — Cube World** and build inside it. Drive the agent with chat: `l` turns left, `r` turns right, `run` starts your solver, `rl` teleports the agent back to you.

**Your maze must include:**

- **2 pistons** (each opens a wall or path when powered)
- **2 doors** (the agent opens them with `agent.interact(...)`)
- **4 AND conditions** in your solution ( `... and ...` )
- **1 OR condition** in your solution ( `... or ...` )

Tick off each part as you go.

## 1 · Design Your 6 Zones

Your maze runs through **6 zones** in a row. Give each zone one job: a turn, a piston, a door, or a junction that checks two things.

**Sketch your 6 zones.** Write each zone number and what makes it tricky.

**Where go your 2 pistons? Which zones, and what does each open?**

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**Where go your 2 doors? What is behind each one?**

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## 2 · Build the Maze

Build it in the world. Walk it yourself first (no code) to check a person can reach the end, then tick each part as you place it.

- ☐ Zone 1 built
- ☐ Zone 2 built
- ☐ Zone 3 built
- ☐ Zone 4 built
- ☐ Zone 5 built
- ☐ Zone 6 built
- ☐ Piston #1 placed and tested
- ☐ Piston #2 placed and tested
- ☐ Door #1 placed
- ☐ Door #2 placed
- ☐ A clear START block and a clear GOAL block

**What broke while you built it? What did you change?**

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```
while True
```

**your** maze.

```
while True:
    agent.move(FORWARD, 1)

    # a junction – turn only when BOTH things are true (AND)
    if agent.detect(BLOCK, LEFT) and agent.detect(REDSTONE, AHEAD):
        agent.turn(LEFT_TURN)

    # a door OR a piston gate – handle either one (OR)
    elif agent.detect(BLOCK, AHEAD) or agent.detect(REDSTONE, RIGHT):
        agent.interact(AHEAD)
        agent.move(FORWARD, 1)
```

# OR .

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## 4 · Count Your Logic

Go back through your solution and **count**. Write the number, then copy one real line as proof.

**How many AND conditions? (need 4)**

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**Copy one AND line:**

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**How many OR conditions? (need 1)**

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**Copy your OR line:**

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**Why did that spot need OR instead of AND? What two things is it checking for?**

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## 5 · Spot and Fix the Bugs

Read each, say what goes wrong, then write the fix.

```
# Should turn left only when a wall is on the left AND redstone is ahead.  
if agent.detect(BLOCK, LEFT) or agent.detect(REDSTONE, AHEAD):  
    agent.turn(LEFT_TURN)
```

**What is wrong, and what is the fixed line?**

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```
# Should open a door ahead, then step through it.  
elif agent.detect(BLOCK, AHEAD):  
    agent.move(FORWARD, 1)  
    agent.interact(AHEAD)
```

**What is wrong, and what is the fixed code?**

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## 6 · Show Your Work 📷 🎥

Type `run` and watch. It sticks — find the spot, fix your code or maze, `rl` to reset, then run until it reaches the **GOAL**.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

**Write your lines here, then say them in your video.**

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Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.